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DTMIR-Pro: Domain Translation with Prompt-based Latent-Space Generalization for Multi-Weather Image Restoration

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Abstract

Multi-weather image restoration seeks to recover scene visibility under rainy, snowy, and hazy conditions, thereby enhancing high-level vision tasks. Existing methods typically train on combined datasets with single-type weather degradations, limiting their generalization to real-world scenarios involving mixed degradations. Domain translation has emerged as a viable solution by generating diverse weather-degraded variants of the same scene. However, current approaches require separate models for each degradation type, resulting in increased system complexity. To address this, we propose DTMIR-Pro, a prompt-based domain translation framework with latent space generalization for multi-weather image restoration. A single trainable network performs multi-domain translation using domain-adaptive prompts and dynamic kernel selection via a proposed Dynamic Multi-Head Attention block to learn diverse degradation patterns. The restoration network takes translated outputs and employs a Multi-Weather Fusion Block with global-local feature streams to capture complex degradations. Furthermore, we introduce a Similarity-Based Encoder Routing mechanism to transfer domain-specific features from the translation encoder to the restoration stage. Extensive experiments on both synthetic and real-world weather-degraded datasets demonstrate the effectiveness and generalizability of the proposed method. The code is made available at https://github.com/AshutoshKulkarni4998/DTMIR-Pro.

1. Introduction

Surveillance applications involve several decisive tasks such as object detection [9], depth estimation [41], semantic segmentation [55], *etc.* Generally, it is assumed that the input data is clean enough for these tasks to maintain significant performance. However, in adverse weather scenarios, these tasks may perform sub-optimally as the input images are severely degraded with rainy, snowy or hazy artefacts, either individually or in a combined manner. Hence, restoring weather affected images is an important pre-requisite. Several approaches are available for generalized weather degraded image restoration through single trainable models [5, 20, 24, 39, 40, 62]. These approaches involved neural architecture search, by considering weather-specific encoders and a generalized decoder [33]. Using such weather-specific encoders increases the overall computational complexity when handling the possi-

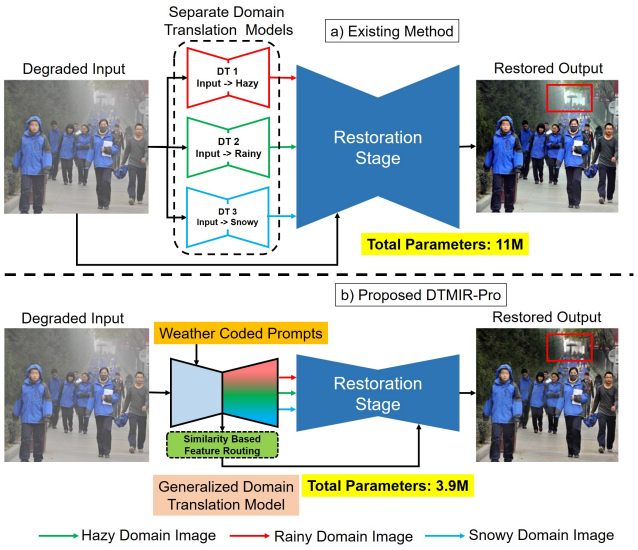


Figure 1. Conceptual difference between the existing domain translation (DT) [40] and the proposed approach for multi-weather image restoration. [40] requires separate models, whereas the proposed method requires only one trainable model for multi-weather domain-translation.

ble weather degradations. Further, there exist transformer based weather specific query learning [20] and knowledge distillation [5] approach through weather specific teacher networks. Recently, an approach with patch-based diffusion model [39] was proposed for multi-weather image restoration. However, these approaches require large number of trainable parameters (KD [5] (28M), TW [20] (31M), WD [39] (110M)). Moreover, reliance of these methods on datasets containing combined images with individual weather degradation limits their performance on real-world weather degraded scenarios.

A potential solution to this in the form of domain translation was recently proposed [40]. **But, how domain translation is helpful?** *Multi-Weather restoration problem is divided into two sub-modules, i.e., domain translation followed by restoration. Domain translation is proposed to produce weather-degraded variants of the input. This makes the restoration network invariant to weather degradation, in turn generalizing it via combined learning of translated weather variants for multi-weather restoration.* We propose a novel prompted generalized multi-domain translation approach for multi-weather image restoration (*3.9M parameters*) (see Fig. 1). A *single trainable network* is designed for multi-domain translation using *weather-coded prompts* that encode discriminative weather-specific cues to enable robust domain translation. To enhance feature extraction, we introduce a dynamic attention block. For restoration, we propose *similarity-based feature routing*, which selects encoder features from the translation network based on the similarity between input and translated images, guiding the restoration process. To handle compound degradations, we introduce a multi-weather feature fusion block that combines features from any two weather types at both global (semantic) and local levels. Compared to existing methods [40], our approach offers two key advantages: (i) a single trainable model replaces the need for N independently trained networks for N degradations, and (ii) similarity-based routing eliminates the dependency on the input image, making the restoration network degradation-invariant. Our main contributions are summarized as:

- The first prompted generalized light-weight multi-domain translation based approach for multi-weather degraded image restoration, having $1/3^{rd}$ parameter (**3.9M**) compared to [40] (11M).
- A novel similarity based feature routing approach for effectively utilizing the encoder features from domain-translation network to guide the multi-weather restoration network.
- A dynamic attention block to extract robust features from input, and a multi-weather feature fusion block to learn combined weather degradations for holistic multi-weather image restoration.

Thorough experimentation on synthetic and real-world weather-degraded datasets evaluates the effectiveness of the proposed method over existing methods for multi-weather image restoration. We also explore the applicability of the proposed method on underwater image enhancement.

2. Related Works

2.1. Weather-specific Image Restoration

De-hazing: Numerous methods have been proposed for image dehazing with [12] and without [18, 23, 25–27] prior information. Yeh *et al.* [54] used structural and statistical feature learning for single-image dehazing. Li *et al.* [31] addressed over/under-dehazing via channel and spatial attention with skip connections. Fan *et al.* [14] proposed a residual attention mechanism with cross-connections using paired depth maps. Recent unpaired learning approaches [43, 53] integrate depth-density priors to improve restoration, though estimating accurate priors in dense haze remains challenging. Furthermore, haze varies widely in the real world, but existing methods often target limited or common haze types, lacking generalizability to diverse unseen scenarios.

De-raining: Rain detection followed by removal has been explored in [52], while Ahn *et al.* [1] introduced attentive gating to balance rain removal. Dilated convolutions with detail recovery were proposed in [11], and Zamir *et al.* [57] developed a progressive multi-stage pipeline. Wang *et al.* [49] used kernel-guided CNNs, and Hu *et al.* [16] incorporated depth to jointly handle rain streaks and fog. Despite these advances, real-world scenarios often involve complex veiling effects, which remain challenging.

De-snowing: Snow removal is particularly difficult due to snow's non-transparent nature. Zheng *et al.* [60] proposed a multi-guided filter, while Wang *et al.* [48] used dictionary-based decomposition. Zhang *et al.* [59] leveraged semantic and depth priors, and Chen *et al.* [4] applied wavelet loss. Recent works [45, 46] explored invertible networks and domain adaptation for improved realism.

While these methods achieve strong results for specific degradations, they fail under complex weather conditions involving multiple factors, such as veiling effects, limiting their effectiveness in real-world multi-weather scenarios.

2.2. Multi-Weather Image Restoration

Earlier research focused on restoration of images with single weather degradation such as haze [28, 44], rain [7, 58] and snow [2, 36] removal. These single weather image restoration approaches perform better for a specific weather degradation, but the dire need of generalized algorithms led the researchers to propose multi-weather image restoration approaches [5, 20, 39, 62]. Valanarusu *et al.* [20] equipped learnable weather queries in the transformer network for decoder generalization. Chen *et al.* [5] distilled

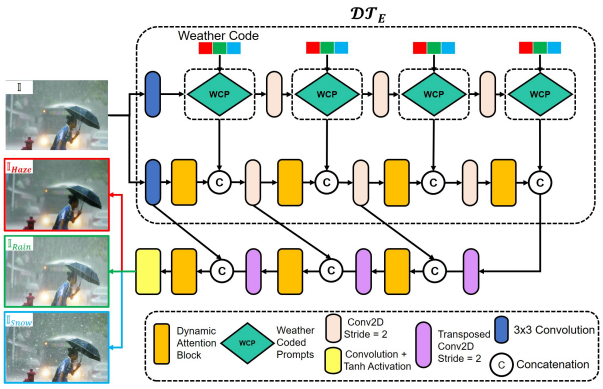


Figure 2. Representation of the proposed prompt-based domain translation network.

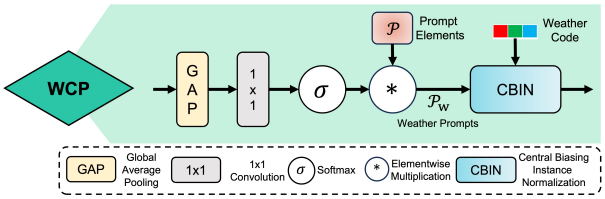


Figure 3. Illustration of the Weather Coded Prompt (WCP) block.

knowledge from weather specific teacher networks to a unified student network. Kulkarni *et al.* [24] utilized two-streamed approach for degradation removal and feature correction. Özdenizci *et al.* [39] proposed a patch based diffusion model for multi-weather image restoration. Recently, [62] trained a network in two stages, where the first stage aims to learn weather specific features, and second stage learns the weather general features. Histoformer [47] employs histogram self-attention and dynamic-range convolution to restore weather-degraded images by grouping similar-intensity pixels. It also uses Pearson correlation loss to preserve linear relationships with ground truth. However, these approaches require computationally complex models for multi-weather image restoration with limited performance on real-world scenarios.

2.3. Domain Translation and Prompt Learning

Various methods have been developed for image-to-image domain translation. Early approaches, such as conditional GANs, CycleGAN [61], and DiscoGAN [22], preserve structural consistency using cycle-consistency loss. StarGAN [10] enables multi-domain translation using a single generator with an auxiliary classifier. A recent method [40] extends this idea for image restoration via multiple independently trained domain translation networks, making the approach computationally expensive and cumbersome. While these methods perform well in one-to-one mappings, they struggle with efficiency and generalization in multi-domain tasks due to the need for multiple generators. SingleGAN [56] addresses this by sharing a single model across domains. Building on this idea, we propose a generalized many-domain translation framework using *weather-coded prompts*, eliminating the need for separate domain-specific networks [40].

Recent advances in visual prompts have shown promise in vision tasks. PromptIR [42] applies prompts in decoder networks for low-level tasks. In contrast, we introduce visual prompts within the domain translation network to generalize across input weather domains. *Key distinctions from [42] are discussed in the supplementary material.*

3. Proposed Approach

In this section, we provide deep insight into the proposed approach for multi-weather image restoration which includes *a) Prompted Generalized Domain Translation* (Figure 2), and *b) Multi-Weather Image Restoration*.

3.1. Prompted Generalized Weather Translation

Existing method [40] trained separate models for different weather-translations, which is a cumbersome process and has higher computational requirements. To avoid this, we propose a *generalized domain translation solution* requiring a single trainable model to translate image from any domain to its respective weather translated domains.

Given an input image $\mathbb{I}$, we obtain its corresponding domain translated images ($\mathbb{I}_{Haze}$, $\mathbb{I}_{Rain}$, and $\mathbb{I}_{Snow}$). For this, we use the prompted encoder based architecture for weather translation as shown in Figure 2. The input (degraded) image is initially passed through a convolution layer to convert the image into features. These features are then further passed through a series of Weather Coded Prompters and downsampling operations to obtain the weather translation specific prompts at various scales. These weather coded prompts are then fused with the successive encoder features to guide the domain translation stage. Further, these encoded features are passed through series of decoder blocks to obtain the domain translated images in accordance to weather coded prompts.

3.1.1. Weather Coded Prompts:

Input images may contain different weather degradations, and translating such inputs to the target domains in a generalized manner requires feature adaptation. For enabling multi-weather translation using a single generator network, learnable weather prompts ($\mathcal{P}_w$) are fused with input features to modulate them in accordance to the weather code (w) provided. Please refer to Figure 3 for the illustration of weather coded prompt block. ***Weather prompts*** are trainable tensors that learn the arbitrary input domain information while optimizing domain translation network. In order to induce target specificity to the prompts, they are injected with weather codes. For easing the input adaptability, we make weather prompts attentively adapt to input weather degradations. This is done by extracting global weights

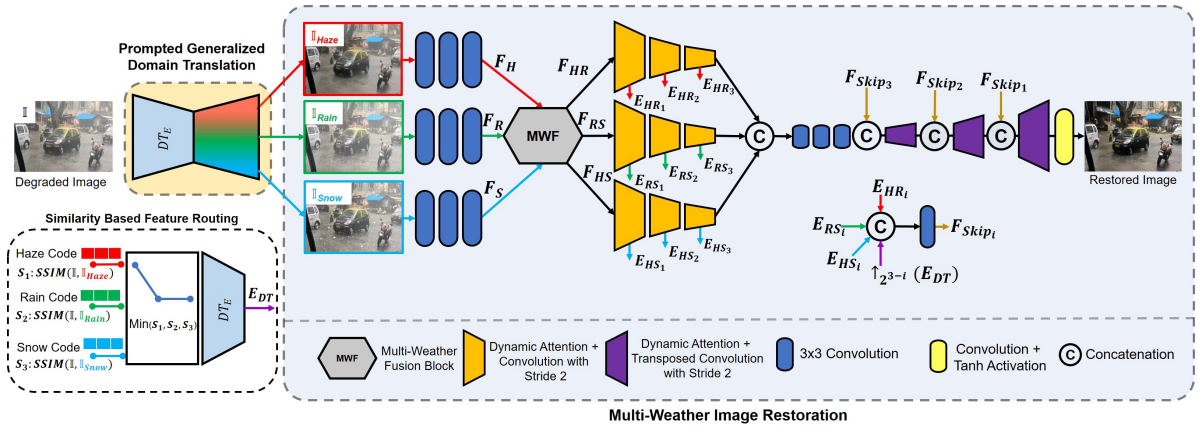


Figure 4. Overview of the proposed DTMIR-Pro architecture for multi-weather image restoration.

from the input features to be multiplied with the weather prompt elements ($\mathcal{P}$) as:

$$\mathcal{P}_w = \sigma\left(C_{1\times1}\left(\mathcal{G}(F_{in})\right)\right) * \mathcal{P}, \tag{1}$$

where σ is softmax activation, $C_{1\times1}$ is convolution with kernel size 1×1, $\mathcal{G}$ is global average pooling, F_{in} are the input features, and $*$ is elementwise multiplication. Weather code is a domain code formulated as a one-hot vector, similar to latent code utilized for denoting the attributes of the generated image. By manipulating this latent code, one can generate new data samples that share similar characteristics to the original data but may differ in certain aspects based on the variations encoded in the latent space. We have set the latent code as: [1, 0, 0] for haze, [0, 1, 0] for rain, and [0, 0, 1] for snow. For injecting these weather codes to the learnable weather prompts, we utilize the central biasing instance normalization ($CBIN$) [56] as:

$$CBIN(\mathcal{P}_w) = \frac{\mathcal{P}_w - E[\mathcal{P}_w]}{\sqrt{Var[\mathcal{P}_w]}} + tanh(f(w)), \tag{2}$$

where f is a learnable affine transformation applied on the weather code w. Such fusion of weather codes and prompts enables target domain specific discriminative feature learning, essential for translating the arbitrary input domain into its corresponding weather domains.

3.1.2. Dynamic Attention Block:

Feature extraction with maximum representation ability and efficiency play a vital role in domain translation. Furthermore, transformer blocks have been proven effective in capturing long range dependencies, yet, traditional transformer architectures use single convolution kernel for query extraction in the attention block. Such setting may restrict the network from extracting diverse features due to static kernels. Dynamic convolutions [8] perform attentive (dynamic) aggregation of parallel convolutions kernels, enabling the diverse feature learning per layer. Hence, we propose the dynamic convolutions ($DynC$) as query (Q_d) extractors in the dynamic attention block as:

$$\mathcal{D}_{Att}(x) = x + \sigma(Q_d \cdot K^T) \cdot V, \tag{3}$$

$$Q_d = ReLU(DynC(x)), \tag{4}$$

$$DynC = \sum_{n=1}^{N} \pi_n(x) W_n, \tag{5}$$

where $K = C_{3\times3}^{depth}(x)$, $V = C_{3\times3}^{depth}(x)$ are keys and values respectively for input features x extracted using depth-wise convolutions with kernel size 3×3 ($C_{3\times3}^{depth}$), $\sigma(\cdot)$ is the softmax activation, $\pi_n(x)$ represents output of squeeze and excitation attention block [15] for generating the feature attention weights in accordance to input x, and W_n, $n \in [1, N]$ represent n^{th} convolution kernel. *More details about dynamic attention block are given in the supplementary material.*

3.2. Multi-Weather Restoration

Upon successful domain translation, the restoration stage takes translated images as input along with outputs from the proposed *similarity based feature routing block* (SBFR) (Sec. 3.2.1). Each domain translated image ($\mathbb{I}_{Haze}, \mathbb{I}_{Rain}, \mathbb{I}_{Snow}$) is processed through convolution layers to extract features F_H, F_R, and F_S. These are fused via the proposed *multi-weather fusion block* (MWF) (Sec. 3.2.2) to obtain joint weather representations (*e.g.*, F_{HR}, F_{RS}, F_{HS}). Each fused feature passes through encoder layers comprising the proposed dynamic attention block (Sec. 3.1) and strided convolution, followed by concatenation.

The aggregated features are forwarded through convolutional and decoder layers, which include dynamic attention and transposed strided convolutions. At each level, MWF features are concatenated with SBFR features (E_{DT}) for enhanced restoration. Skip connections $F_{Skip_i}, i \in (1, 2, 3)$ are defined as:

$$F_{Skip_i} = \text{Conv}_{3\times3}(\langle E_{HR_i}, E_{RS_i}, E_{HS_i}, \uparrow_{2^{3-i}}(E_{DT})\rangle),$$

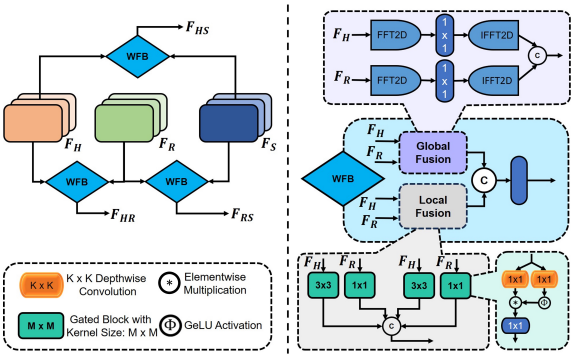


Figure 5. Schematic of the proposed multi-weather fusion block consisting of weather fusion block (WFB) to fuse pairs of weather degradations.

where $\uparrow_{2^n}$ denotes transposed convolution with stride 2^n, and $\langle\cdot\rangle$ indicates feature concatenation. The final restored image is obtained by passing the decoder output through a 3-channel convolution layer.

3.2.1. Similarity Based Feature Routing

The domain translation network learns mappings from input to target weather domains using weather-coded prompts. *Qualitative analysis (Fig. 7) and quantitative results (in supplementary material) reveal that the learned features effectively suppress the input degradation while translating to the target domain.* We leverage these prompted features in the restoration network to enhance performance.

To avoid redundancy from processing the input image, as in [40], we propose *similarity-based feature routing*, which guides the restoration network using encoder features from the translation network. Given an input degraded image $\mathbb{I}$ and its domain-translated variants ($\mathbb{I}_{Haze}$, $\mathbb{I}_{Rain}$, $\mathbb{I}_{Snow}$), we compute the structural similarity (SSIM) between $\mathbb{I}$ and each translated image. The translation yielding the *minimum similarity* is selected, implicitly indicating the dominant degradation in $\mathbb{I}$.

For example, if $\mathbb{I}$ is a snowy image, its SSIM is typically lowest with the hazy variant, i.e., $\mathbb{I}_{Haze}$. Thus, features from the haze translation path (representing "snow removal–haze addition") are routed to the restoration network. Let $\mathcal{DT}_E$ denote the last encoder layer of the domain translation network; the routed features $SBFR_O$ are computed as:

$$S_i = SSIM(Y(\mathbb{I}), Y(\mathbb{I}_i)), \tag{6}$$

$$j = Index(min([S_i])), i \in \{Rain, Haze, Snow\}, \tag{7}$$

$$SBFR_O = \mathcal{DT}_E(\mathbb{I}, w_j), \tag{8}$$

where Y depicts Y-Channel output obtained after converting RGB images into YCbCr images. This conversion is done because Y channel of an image contains discriminative degradation information [13]. Therefore with the similarity based feature routing, the encoder features of weather translation network act as prompts themselves, for guiding the restoration network in accordance to input. Unlike [40], this further eliminates the need to use the actual input image in the restoration network, as we already have the domain-translated images and the similarity-based routed features.

3.2.2. Multi-Weather Feature Fusion

In real world scenarios, the images captured in adverse weather conditions may get affected by combined occurrence of multiple weather degradations (such as rain + veil, snow + veil, *etc*). Hence, it is vital to effectively fuse the individual weather features to learn their combined effect. To achieve this, we propose a multi-weather feature fusion module (MWFF). The schematic of MWFF is presented in Figure 5. The main aim of MWFF is to fuse any pair of weather degradations, both in global (semantic) level and local level. The proposed MWFF consists of dedicated blocks for global and local level fusion. In global level fusion, the two weather features are passed through $\texttt{FFT2d} \rightarrow \texttt{Conv}_{1\times1} \rightarrow \texttt{IFFT2d}$ to capture global receptive field. Further, for capturing the various sizes of weather degradations (such as varying rain and snow sizes), and to fuse information on a local-level, multi-scale gated depthwise convolutions are used (please see Figure 5). The multi-scale gated depthwise convolutions are equipped with different kernel sizes. Such global and local fusion of features enables the restoration network to learn various combinations of weather degradations. *The effectiveness of each proposed block is verified in ablation study (Sec. 4.5).*

4. Experiments and Analysis

4.1. Training Details

We follow the training datasets used in [40] for training the domain translation network, except the fact that [40] trains separate networks for each domain translation, we train our domain translation network in a unified manner. After training the domain translation network, we freeze its weights, and use its outputs for training the restoration network. The domain translation network is trained for 100 epochs and restoration network is trained for 300 epochs. While training the translation and restoration networks, the initial learning rate is set to 10^{-3} which is varied using cosine annealing strategy. The weights of both the networks are updated on NVIDIA-GeForce GTX 1080 8GB GPU. The discriminator network is same as in [56].

Domain Translation Loss Functions: The aim of the domain translation network ($\mathcal{DT}$) is learning mapping from input domain ($\mathbb{I}$) to corresponding target domains $\mathbb{I}_i, i \in \{Rain, Haze, Snow\}$ given a weather code (w). The domain translation network is trained adversarially having discriminator networks $D_i, i \in \{Rain, Haze, Snow\}$ with



Figure 6. Qualitative results comparison of the proposed (Ours) and existing UMVR [24], TW [20], KD [5], WGWS [62], WD [39], DMIR [40] methods on real-world haze, rain, and snow degraded images (*please zoom in for clearer view*).

Table 1. Quantitative results analysis on real-world datasets in terms of NIQE [37] (↓), Entropy [6] (↑), and BRISQUE [38] (↓). Best and second best values are denoted in **Red** and Blue.

Dataset	Methods	NIQE	Entropy	BRISQUE
RTTS	UMVR [24]	5.009	7.221	28.625
	KD [5]	4.996	7.297	26.837
	TW [20]	5.703	7.263	29.874
	WGWS [62]	6.199	7.064	35.715
	WD [39]	5.315	7.115	29.897
	PIR [42]	5.127	7.342	27.593
	Histoformer [47]	4.923	7.428	25.847
	DMIR [40]	4.859	7.505	24.761
	DTMIR-Pro	**3.599**	**7.551**	**23.951**
RID	UMVR [24]	6.604	7.486	24.296
	KD [5]	6.943	7.459	24.841
	TW [20]	7.496	7.393	24.165
	WGWS [62]	6.813	7.348	24.862
	WD [39]	7.103	7.401	24.021
	PIR [42]	7.284	7.437	24.315
	Histoformer [47]	6.871	7.463	24.128
	DMIR [40]	7.625	7.492	23.931
	DTMIR-Pro	**6.252**	**7.527**	**23.397**
Snow-Realistic	UMVR [24]	4.296	7.354	23.761
	KD [5]	6.643	7.459	24.841
	TW [20]	5.628	7.331	25.377
	WGWS [62]	4.417	7.241	25.331
	WD [39]	4.342	7.324	23.996
	PIR [42]	4.573	7.398	22.847
	Histoformer [47]	4.285	7.541	22.163
	DMIR [40]	4.196	7.572	21.884
	DTMIR-Pro	**3.668**	**7.617**	**20.125**

the training objective:

$$\mathcal{L}_{DT_{adv}} = \sum_{i} \mathbb{E}[log(D_i(\mathbb{I}_i))] + \mathbb{E}[log(1 - D_i(\mathcal{DT}(\mathbb{I}, w_i)))]. \tag{9}$$

We calculate the contrastive loss [51] ($\mathcal{CL}_{DT}$) between the domain translated images. By enforcing a larger inter-distance between the translations, the contrastive loss helps ensure that the translated images retain distinct visual features associated with their respective weather conditions. In the context of weather degradations, this means preserving the semantic differences specific to each weather condition, such as haze, rain, or snow. This is done by considering the domain translated image as anchor, its weather degraded ground-truth as positive image, and the images from other translated domain as negatives. The total domain translation loss is defined as:

$$\mathbb{L}_{DT_{Total}} = \mathcal{L}_{DT} * \lambda_{DT_1} + \mathcal{CL}_{DT} * \lambda_{DT_2}, \tag{10}$$

with $\lambda_{DT_1} = 0.5$ and $\lambda_{DT_2} = 1.0$ as weights assigned to domain translation losses.

Multi-Weather Restoration Loss Functions: For multi-weather image restoration network, we utilize the L_1 loss between the restored image and corresponding restoration ground truth. Further, we also calculate the contrastive loss ($\mathcal{CL}_R$) for obtaining perceptually clearer outputs considering the restored image as anchor, the domain translated images ($\mathbb{I}_{Haze}, \mathbb{I}_{Rain}, \mathbb{I}_{Snow}$) as negatives and the restoration ground truth as positive. Hence, the total restoration loss ($\mathbb{L}_R$) is calculated as:

$$\mathbb{L}_R = L_1 * \lambda_{R_1} + \mathcal{CL}_R * \lambda_{R_2}, \tag{11}$$

where $\lambda_{R_1} = 1.0, \lambda_{R_2} = 0.75$ are weights assigned to the image restoration losses. *Detailed equations of the loss functions are given in supplementary material.*

4.2. Datasets

1) SOTS Dataset [29]: We consider 5000 images from outdoor-haze train split and 500 outdoor images from test split for training and testing respectively.

2) Outdoor Rain Dataset [32]: Comprising rainy images

Table 2. Quantitative results on synthetic datasets: SOTS (for haze removal), CSD (for snow+haze removal), and ORD (for rain+haze removal) in terms of PSNR [21]/SSIM [50].

Methods	SOTS	CSD	ORD
UMVR [24]	33.41/0.980	28.65/0.900	22.99/0.830
KD [5]	34.64/0.985	31.35/0.950	29.05/0.916
WD [39]	28.00/0.966	26.85/0.935	29.72/0.921
TW [20]	32.45/0.955	29.76/0.940	27.96/0.950
WGWS [62]	31.31/0.970	28.42/0.925	25.31/0.900
PIR [42]	30.58/0.974	28.95/0.938	30.80/0.923
DMIR [40]	36.26/**0.987**	32.95/0.942	31.24/0.951
Histoformer [47]	33.15/0.962	30.48/0.928	32.08/0.938
DTMIR-Pro	**36.92**/0.981	**33.94**/**0.953**	**32.88**/**0.954**

with veiling effect, 9000 and 750 images from this dataset are used for training and testing respectively.

3) Comprehensive Snow Dataset [3]: 8000 and 2000 images are used for training and testing, respectively, with a combination of snow and fog degradation.

Moreover, performance on real-world scenarios is evaluated using Real-world Task-driven Testing Set (RTTS) [29], Rain In Driving (RID) [34], and Snow-Realistic datasets [36] containing hazy, rainy, and snowy images respectively.

4.3. Quantitative Results

For performance evaluation, we compare results of the proposed method quantitatively on both synthetic and real-world datasets. We compare quantitative results on real-world datasets in terms of reference-free metrics such as Naturalness Image Quality Evaluator (NIQE) [37] (↓), Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE) [38] (↓) and Entropy [6] (↑) (↓ - *lower is better,* ↑ - *higher is better*) in Table 1. The quantitative analysis on synthetic datasets is provided in Table 2. For synthetic datasets, we use the peak signal-to-noise ratio (PSNR) [21] and structural similarity index (SSIM) [50] as evaluation metrics. As seen from the results, the proposed method outperforms the existing state-of-the-art methods.

Parameter Analysis: Being a pre-processing step for higher-level computer vision applications, image restoration is expected to have low-computational overhead. Comparison with existing methods in terms of trainable parameters, model size, and floating point operations (GFLOPs) on image size 256×256 is given in Table 3. As seen from the values, the proposed method has lesser computational footprint as compared to existing methods.

4.4. Qualitative Results

Restoration Results: For visual analysis, we compare the results of the proposed method with the existing state-of-the-art methods (UMVR [24], KD [5], TW [20], WGWS [62], WD [39], DMIR [40]), on real-world datasets for multi-weather image restoration in Figure 6. *More visual results on synthetic and real-world images are provided in the supplementary material.* As seen from the results, the proposed method better enhances the perceptual quality of the image. For hazy images, the proposed method removes more distant (in-depth) haze as compared to existing methods. For rainy and snowy images, the proposed method is able to remove more rain streaks and snow-streaks along with veiling effect. For cross domain analysis, we have also provided some results in night-time conditions in the *supplementary material* by testing the proposed model directly (without retraining for night-time restoration). This analysis verifies the generalizability of the proposed method across various weather degradations with consistently effective restoration performance.

We have provided qualitative results of domain translation in the supplementary material.

Table 3. Comparison based on number of parameters, model size, and GFLOPs on image size 256×256. Our parameters are given by considering the domain translation and restoration networks.

Methods	Par.(M)	Size (MB)	GFLOPs
KD [5]	28M	348.89	49.22
TW [20]	31M	148.77	**12.24**
WD [39]	110M	1296.71	475.43
PIR [42]	35.5M	397.51	158.15
DMIR [40]	11M	44.01	100.14
DTMIR-Pro	**3.9M**	**15.76**	28.60

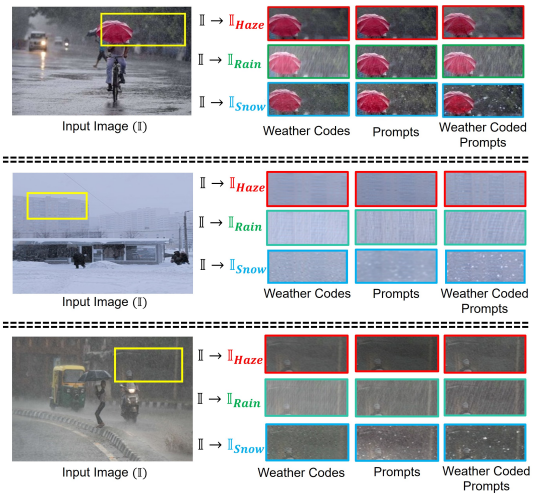


Figure 7. Ablation study visual results with different types of domain translation settings (*please zoom in for clearer view*).

4.5. Ablation Studies

To evaluate the effectiveness of the proposed method, we perform various ablation studies elaborated in this section. We provide the performance variations on SOTS (**H**), ORD (**R**) and CSD (**S**) datasets in terms of PSNR.

Weather Coded Prompts: We evaluate *whether weather coded prompts help in effective input to domain translation as compared to just domain codes or prompts?* This is verified by comparing results when the latent space is manipu-

Table 4. Ablation study results in terms of PSNR on various ablation settings. H: SOTS, R: Outdoor Rain, S: CSD Dataset.

	Setting	PSNR-H-R-S
SBFR Ablation	No Feature Routing	35.81/31.27/32.24
	Direct Feature Routing	36.24/31.95/33.12
	SBFR	**36.92/32.88/33.94**
Attention Block Ablation	Convolution Attention	35.53/30.39/32.12
	Depthwise Conv. Attention	35.95/31.76/32.89
	Dynamic Attention	**36.92/32.88/33.94**
Multi-Weather Fusion Ablation	No Fusion	35.47/30.86/31.72
	Additive Fusion	35.68/31.12/32.18
	Concatenation	35.84/31.73/32.63
	Global Fusion	36.33/32.21/ 32.78
	Local Fusion	36.51/32.45/33.12
	MWF	**36.92/32.88/33.94**

lated using only domain codes, prompts, and weather coded prompts. The visual results are provided in Figure 7. As seen from the results, the domain translated images generated using the proposed weather coded prompts have better diversity and realistic-looking degradations when compared to other settings.

Similarity Based Feature Routing (SBFR): SBFR is proposed to utilize domain translation features in restoration network to provide input domain guidance and avoid redundant feature processing. But, *is the proposed similarity based feature routing really effective for multi-weather image restoration?* Experimentation results reported in Table 4 verify that proposed SBFR performs better as compared to no routing (*viz.* restoration only using the domain translated images), and direct routing (*viz.* restoration using the encoder features obtained using "all-weather" domain code). *Another ablation study on feature routing is provided in supplementary material.*

Dynamic Attention Block (DAB): DAB equipped with dynamic kernels is proposed to learn diverse attentive features from the inputs. To analyse the performance of different attention that can be utilized in encoders and decoders, we perform various experiments. We use the traditional convolution attention, depthwise convolution based attention, and the proposed dynamic attention. The results in Table 4 suggest that the proposed dynamic attention achieves better restoration performance.

Multi-Weather Feature Fusion (MWF): MWF aims at learning combinations of weather degradations beneficial for image restoration in real-world scenarios. To scrutinize *whether the proposed MWF block is really favourable towards effective restoration*, we perform different experiments. These experiments involve no feature fusion (direct encoding of F_H, F_R, and F_S), additive fusion, concatenation based fusion, local fusion, global fusion, and proposed MWF. Results in Table 4 verify that the fusion of multi-weather features using the proposed MWF turns out to be more effective. *Please see the supplementary material for more ablation results.*

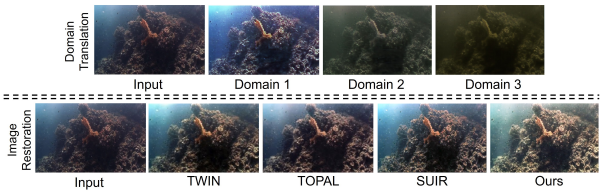


Figure 8. Underwater image enhancement analysis. *Top Row*: Domain translation results of the proposed method, *Bottom Row*: Image restoration results comparison of the proposed method with existing TWIN [35], TOPAL [19], and SUIR [17] methods.

Table 5. Quantitative results on UIEB dataset for underwater image enhancement.

Methods	TOPAL [19]	TWIN [35]	SUIR [17]	Ours
PSNR	21.95	22.30	24.59	**25.17**
SSIM	0.902	0.880	0.901	**0.918**

5. Applications of the Proposed Framework

Underwater Image Enhancement: Underwater images exhibit diverse color degradations, making domain translation a feasible solution for restoration. We train a domain translation network to convert colored underwater images into varicolored variants, followed by a restoration network trained on the UIEB dataset [30]. Qualitative results are shown in Figure 8, and quantitative results on UIEB in Table 5. The proposed method outperforms existing approaches, highlighting its potential for broader restoration tasks.

Restoration capability of any algorithm is utilized in many higher-level computer vision tasks as a preprocessing step. *We have provided visual results on various such computer vision applications in the supplementary material.*

6. Conclusion

In this paper, we introduced a novel domain translation with prompt based latent space generalization approach for multi-weather image restoration. We proposed a novel prompt based domain translation network for converting the input image from any arbitrary weather domain to its respective domain translated representatives. Dynamic attention block is proposed for capturing diverse features for effective domain translation and restoration. Further, we introduced a novel similarity based feature routing approach to collaborate the domain translation network with restoration network. Finally, multi-weather fusion block is proposed containing global fusion and local fusion branches to learn mixed-weather degradations. Substantial experiments on synthetic as well as real-world datasets demonstrated the effectiveness of the proposed DTMIR-Pro for multi-weather image restoration. We further demonstrated the applicability of our approach on different applications.

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